

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()
image_name = list(uploaded.keys())[0]

# Read image
img = cv2.imread(image_name, cv2.IMREAD_GRAYSCALE)

# -----
# Pixel Resolution
# -----
low_pixel = cv2.resize(img, (128, 128))
low_pixel = cv2.resize(low_pixel, (img.shape[1], img.shape[0]),
                        interpolation=cv2.INTER_NEAREST)

# -----
# Intensity Resolution (4-bit)
# -----
low_intensity = (img // 16) * 16

# -----
# Display Results
# -----
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(low_pixel, cmap='gray')
plt.title("Low Pixel Resolution")
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(low_intensity, cmap='gray')
plt.title("Low Intensity Resolution")
plt.axis('off')

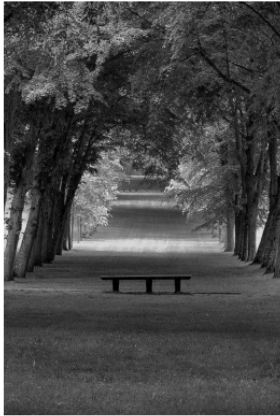
plt.show()
```

[Choose Files](#) NATURE.jpg

NATURE.jpg(image/jpeg) - 111130 bytes, last modified: 7/27/2026 - 100% done

Saving NATURE.jpg to NATURE.jpg

Original Image



Low Pixel Resolution



Low Intensity Resolution

